



Thermal imaging at work on head count in a retail outlet (Credit: <https://behavioranalyticsretail.com>)

Conventional card-based loyalty schemes can incentivise consumers but are not very helpful in impacting customer behaviour. Rather, stores can use NFC tech to develop a frictionless and engaging loyalty scheme that motivates customers to be more loyal.

British sushi chain Wasabi used Thyngs technology to create an app-free loyalty scheme that lets customers tap a smart card with their phone to join Wasabi Club and collect loyalty points in-store.

NFC technology can also be used as a simple and novel way for retailers to get immediate customer opinion. Shoppers can be asked to provide service feedback by tapping NFC chips on lanyards worn by in-store staff. UK-based Asian fast-food chain Wagamama used Thyngs-enabled smart cards when trying out its new MA MA menu at its South Bank branch in London. Wannabe tasters were able to test some of the new menu items on complimentary basis and then rate their dish by just tapping their phone on a smart card.

Thyngs platform then lets retail stores to monitor any such engagement in real time, supplying comprehensive analytics on how effective their marketing strategies are. Such

insights are generally not viable for offline marketing campaigns.

Smart shopping carts

Smart shopping devices that are likely to be widely used in the near future are being developed by technology leaders including IBM, NCR, Fujitsu and Hewlett-Packard. One such device is a supermarket smart cart that has a screen embedded in its handles. A shopper simply needs to swipe a smart card through a slot in the cart to trigger the device on. The screen then communicates with the customer and alerts him or her of various in-store deals and promotions.

Data with respect to customer purchase history is also maintained, and this enables the cart to anticipate which direction the customer is likely to move on the retail shop floor. Customised suggestions then begin to pop up on screen, to help and aid the shopper. The cart maintains a track of all products kept inside it. This means much shorter queues at the billing section as the cart can communicate with the cash register.

Thermal imaging

Thermal imaging cameras operate in the infrared (IR) range of the electro-

magnetic spectrum (roughly 9000 to 14,000 nanometres, or 9µm to 14µm) and generate images of that radiation, known as thermograms. Every object emits IR radiation, which is more than absolute zero, and quantum of radiation emitted by an object goes up as its temperature increases.

It is a well-known fact that temperature of human beings is higher than most objects in the environment. On this basis, thermography is used to detect and count the number of humans. Large retail stores can deploy array sensors that detect heat sources in thermal imaging systems.

These systems are generally based on embedded technology and are placed at a height for precise results. Since these systems detect heat emitted by people, they are able to count well even in varying intensity of lighting. Moreover, they do not need to employ complex background removal algorithms, which is a pre-requisite for computer vision systems. This results in a more precise individual human count. However, such systems are likely to fail in cold weather when customers are wearing thick woollens that help in insulating their body heat.

People counters help in generating actionable customer insights and analytics from such useful data points as footfall, path customers take when moving inside a store, time of the day customers enter the store and so on.

This technology can also help in measuring customer reactions to in-store promotions as well as price changes, and also make linkages between footfall and sales. Information and insights ultimately contribute towards improving store layouts and quality of service.

SixthSense technology

Indian computer scientist and inventor Pranav Mistry's videos wherein he showcases SixthSense technology are very popular. Developed by MIT Media Lab's Fluid Interfaces Group, the tech is a wearable gestural interface that augments the physical world around us with digital information. The prototype is a combination of a computer and cellphone. It encompasses a pocket projector, mirror and camera. Hardware elements are integrated into a pendant-like mobile wearable